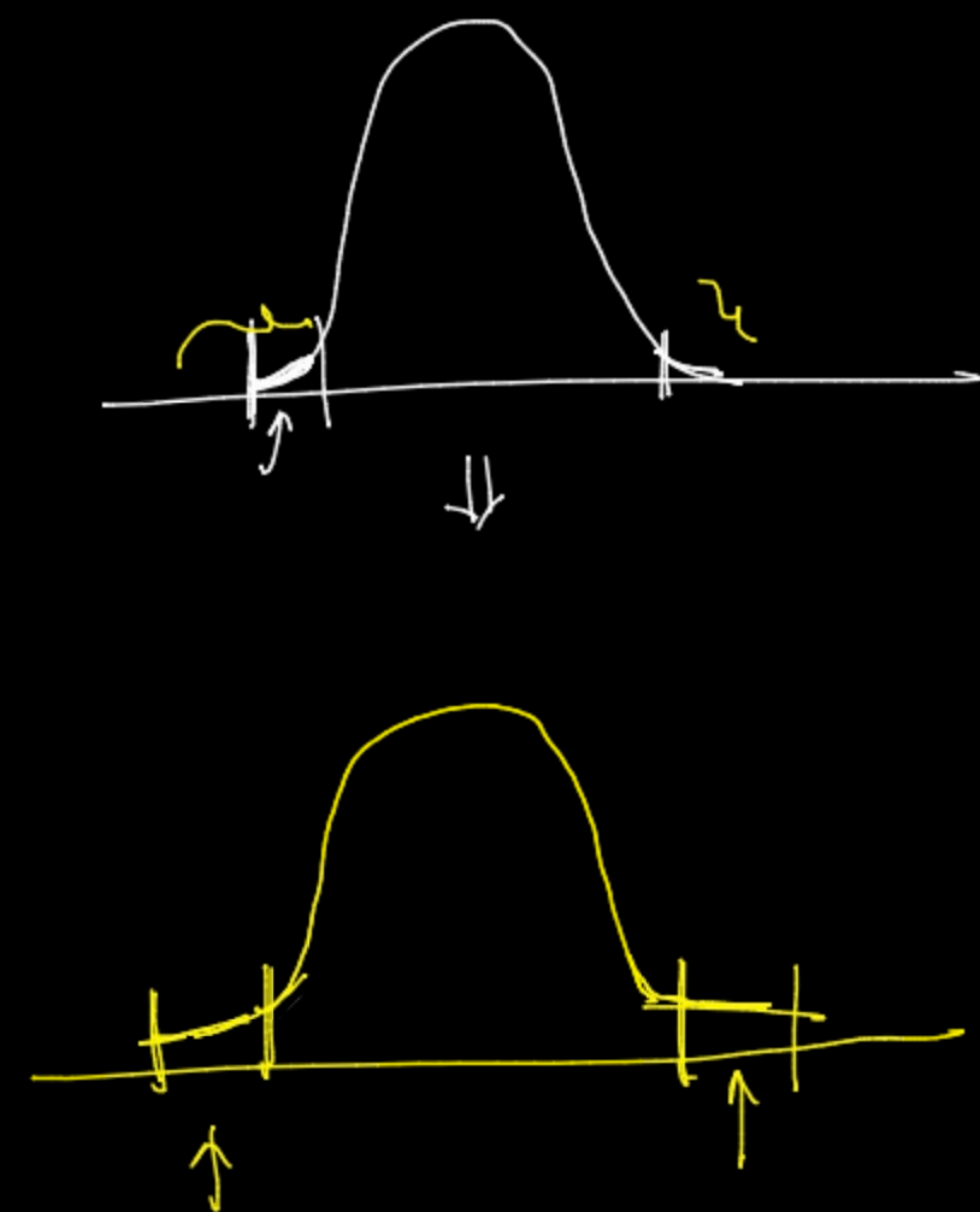
Log Normal distribution

$X \sim N(\mu, \sigma)$

$x = \{x_1\ x_2\ x_3\ ----\ x_n\}$

$\log(x) = \{\log(x_1), \log(x_2) - -\}$



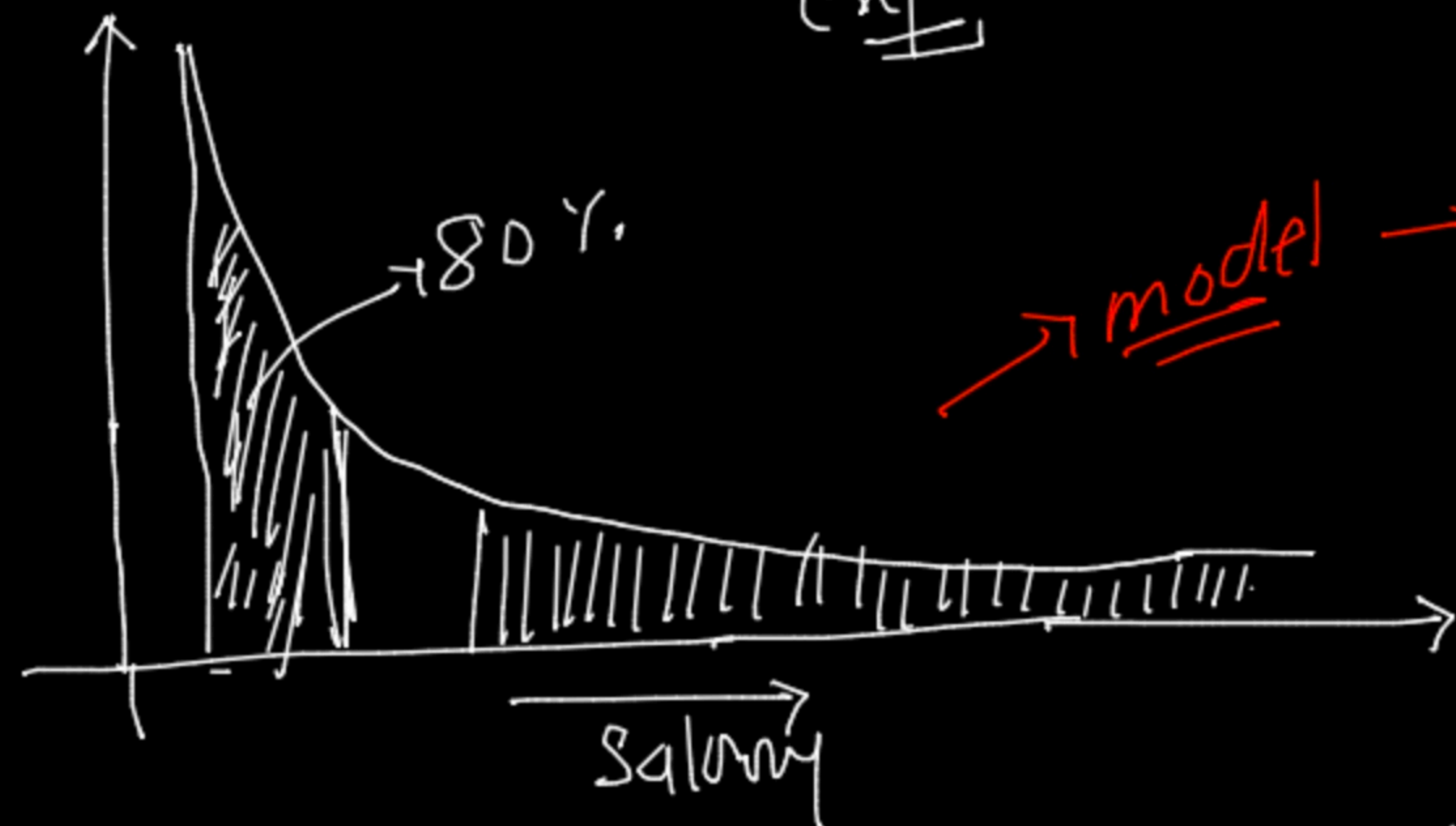
log

Power law

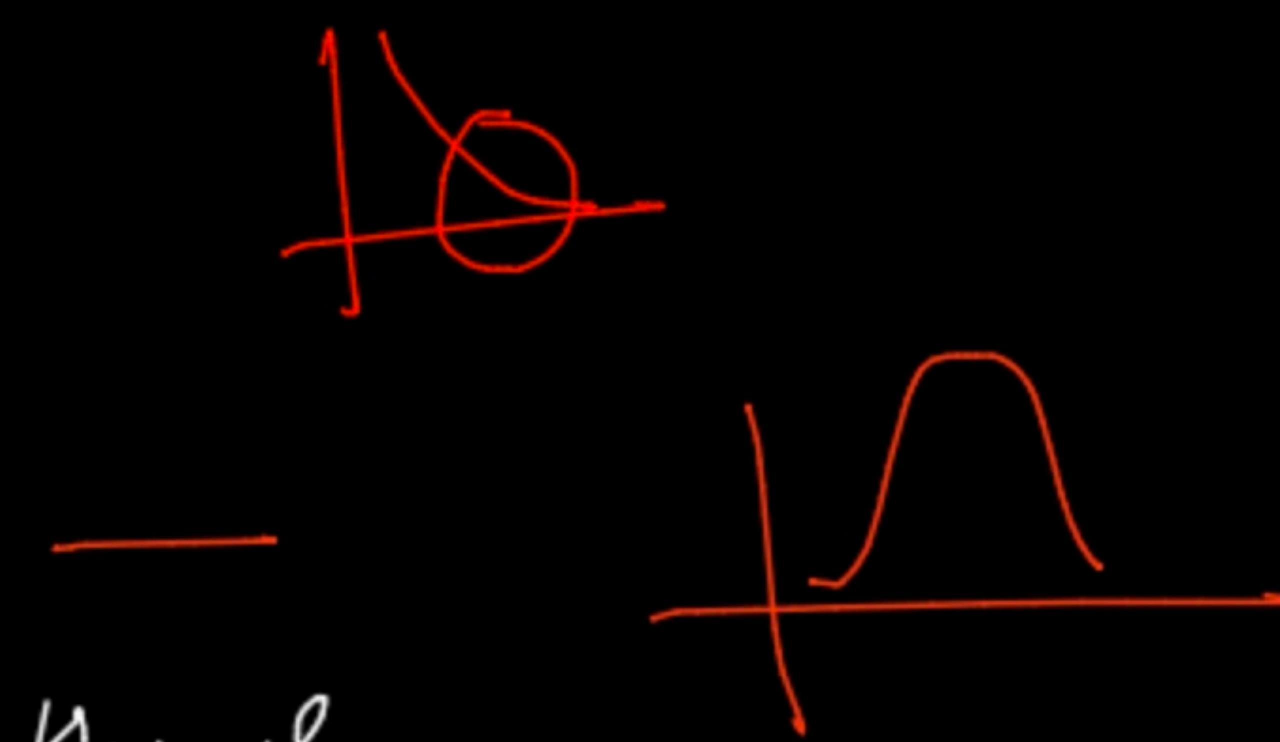
(80-20 Rule)

Var(x)

expl01



→ model → struggle → learn



Pareto distribution

80% time you will find 20% popular things

$\boxed{h.D}$

model → learn

2 / 3

You are screen sharing

Stop share

Power-transform

$X \xrightarrow{P-T} ND$

'Modelling'

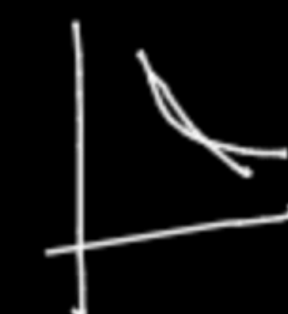
$\text{Box-cox}(x) = \text{lambda}(x)$

scipy.stats.boxcox

scipy | sklearn

implementation

pareto distribution



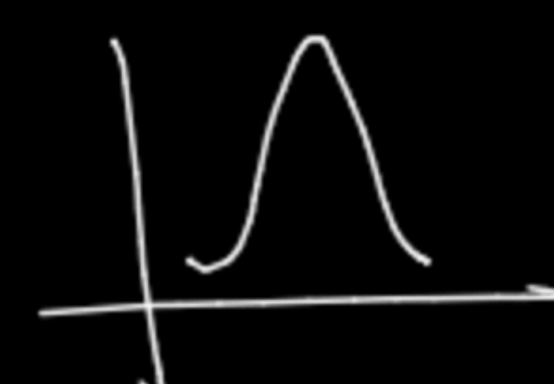
$X = \{12, 14, 15 - - - \}$

$X' = \{-, -, - - - - \}$

x

transform

plot



$Y_i = \frac{x_i^{\lambda} - 1}{\lambda}$ if $\lambda \neq 0$

$= \log(x_i)$ if $\lambda = 0$

assum

$\boxed{\lambda = 2}$

$\cancel{x}$ $\boxed{ND}$

$\frac{12^2 - 1}{2} \Rightarrow \frac{145 - 1}{2} \Rightarrow \frac{143}{2} \rightarrow$ -

Co-variance

X Y

weight height

Q) Is there any Relationship b/w (X & Y)

ie. ✓ X↑ → Y↑ ? wht↑ → w↓,↑

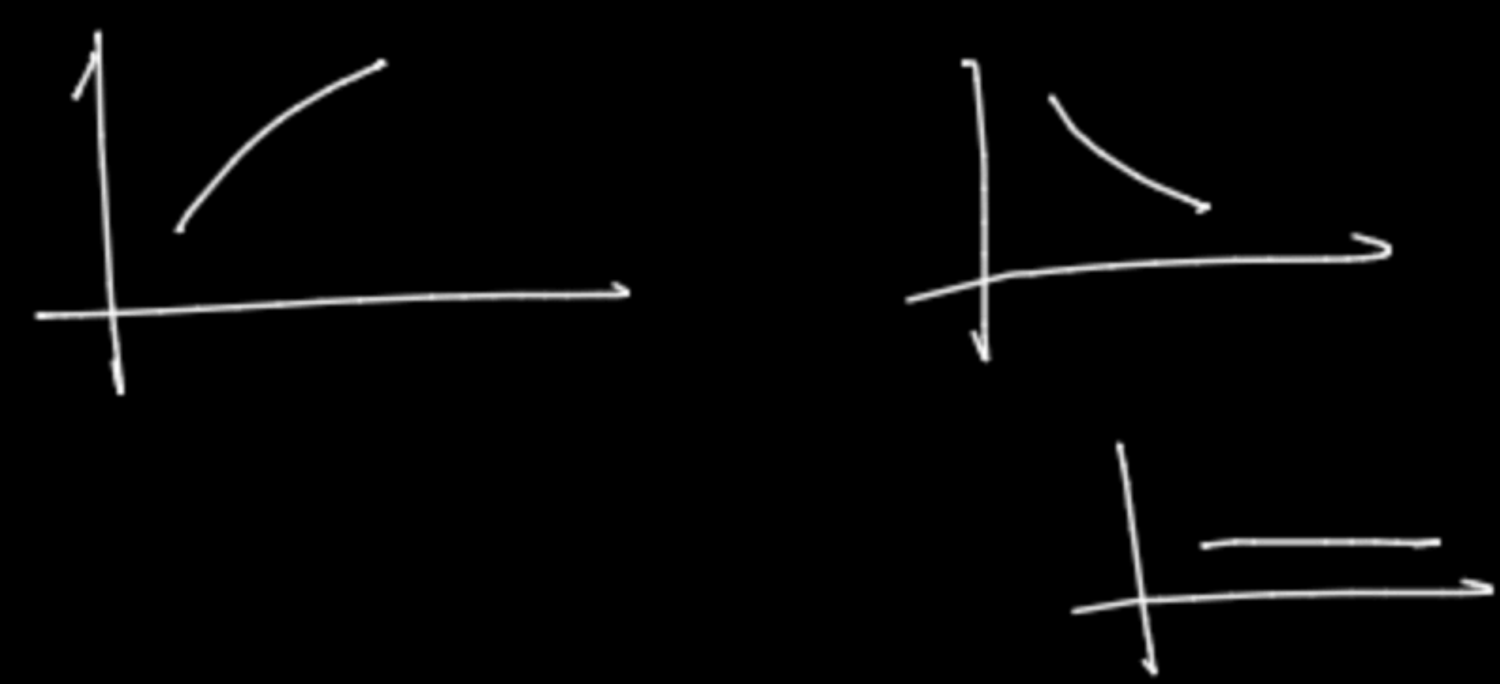
✓ X↑ → Y↓ → model

conclusion

'co-variance'

✓ ① 'co-variance'

✓ ② ↳ Persons correlation coeffⁿ ③ spearman correlation coeffⁿ



(10-20 features)

Best '100' X X

Dx EB □ + Best learnings

analysis → 'input' → ?

$$\operatorname{cov}(X, Y) = \frac{1}{n} \sum_{i=1}^{n} (x_i - \mu_x) * (y_i - \mu_y)$$

mean of x

mean of y

$$\operatorname{var}(X, X)$$

$\boxed{\mu = X1}$ $\mu_y -$

x	y	x - μi	y - μi

mean(y)

mean(x)

μ_x

μ_y

Add → $\overline{n}$

$$\operatorname{var}(X) = \frac{1}{n} \sum_{i=1}^{n} (x_i - \mu_x)(x_i - \mu_x)$$

$$\boxed{\operatorname{cov}(X, X) = \operatorname{var}(X)}$$

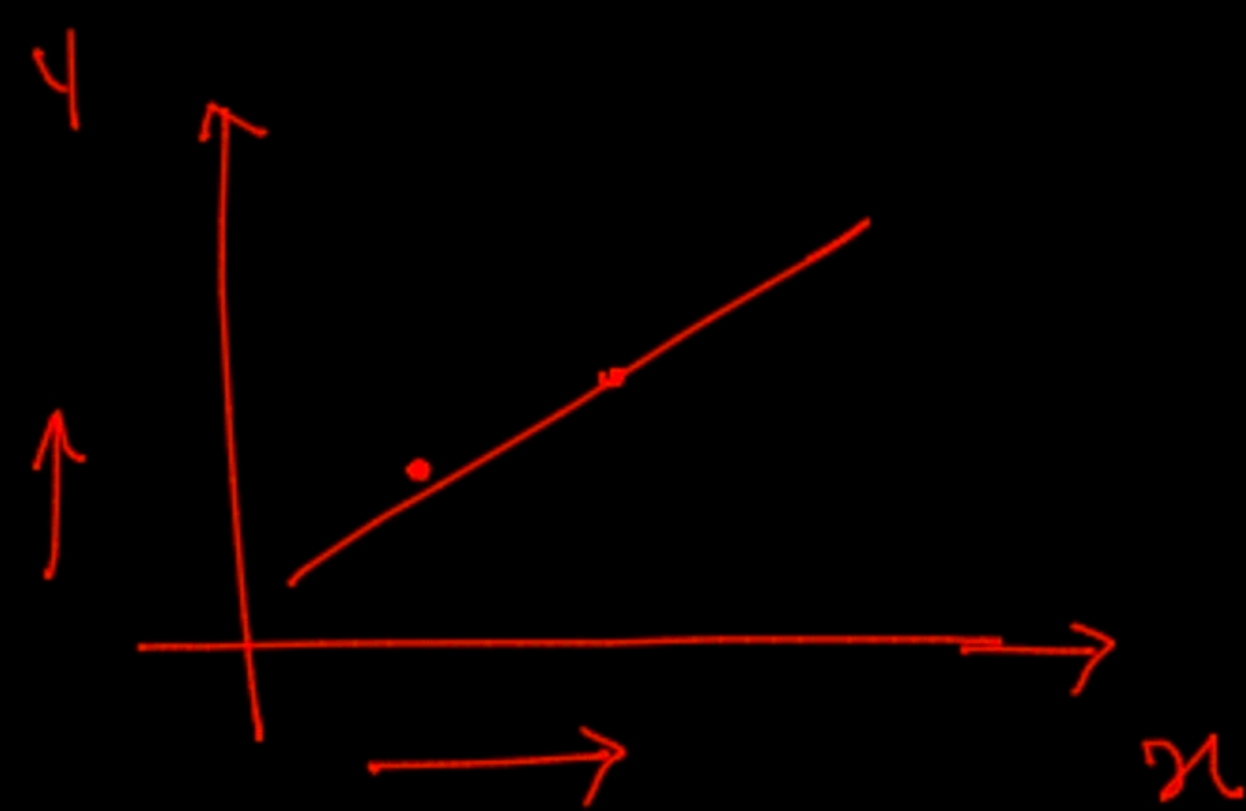
X X

if $\begin{cases} cov(X,Y) = +ve & \text{then } x\uparrow \ y\uparrow \\ cov(X,Y) = -ve & \text{then } x\uparrow \ y\downarrow \end{cases}$

math/Number

(+) (-)

case i

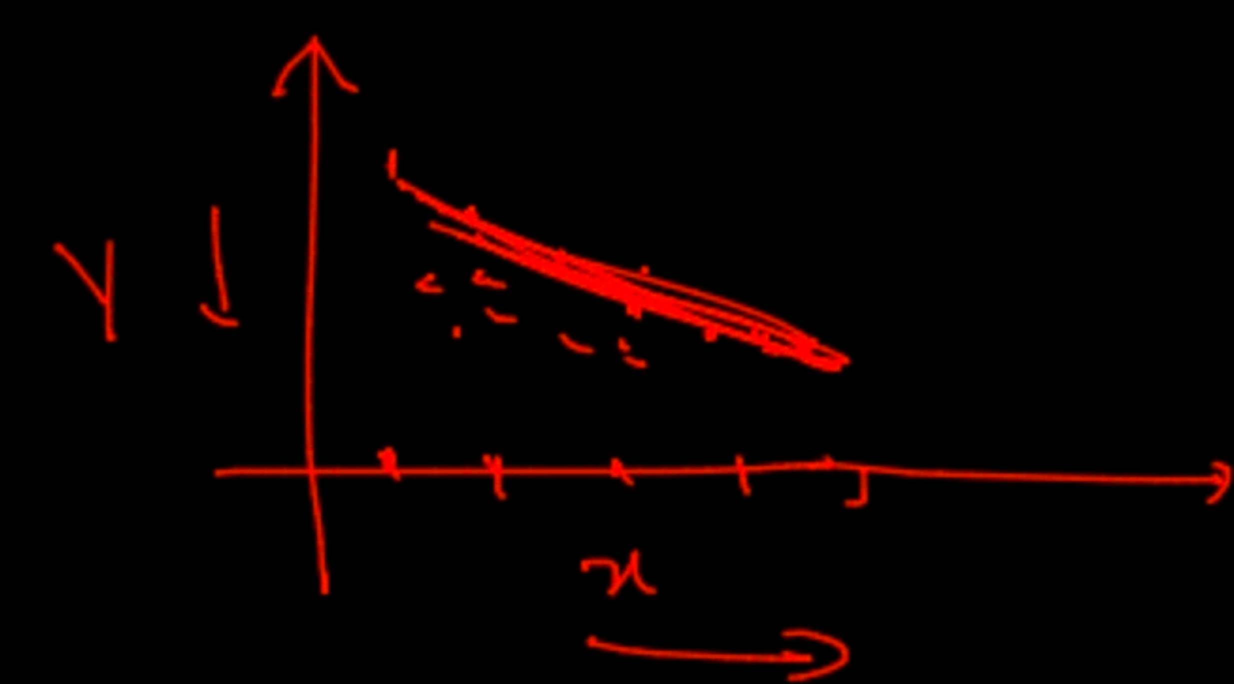


x

↓

y is also

+ +



20% → Y↑ ↓

② Pearson co-relation coeff

⇓

(-1 to 1)

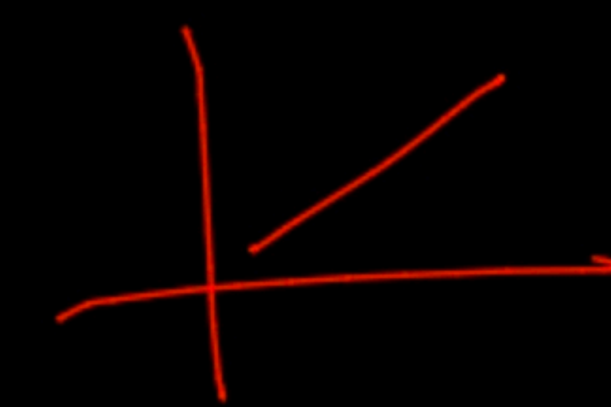
(0.9)

$$\rho_{X,Y} = \frac{cov(X,Y)}{\sigma_X \sigma_Y}$$

-1 → 1

0.5

heatmap

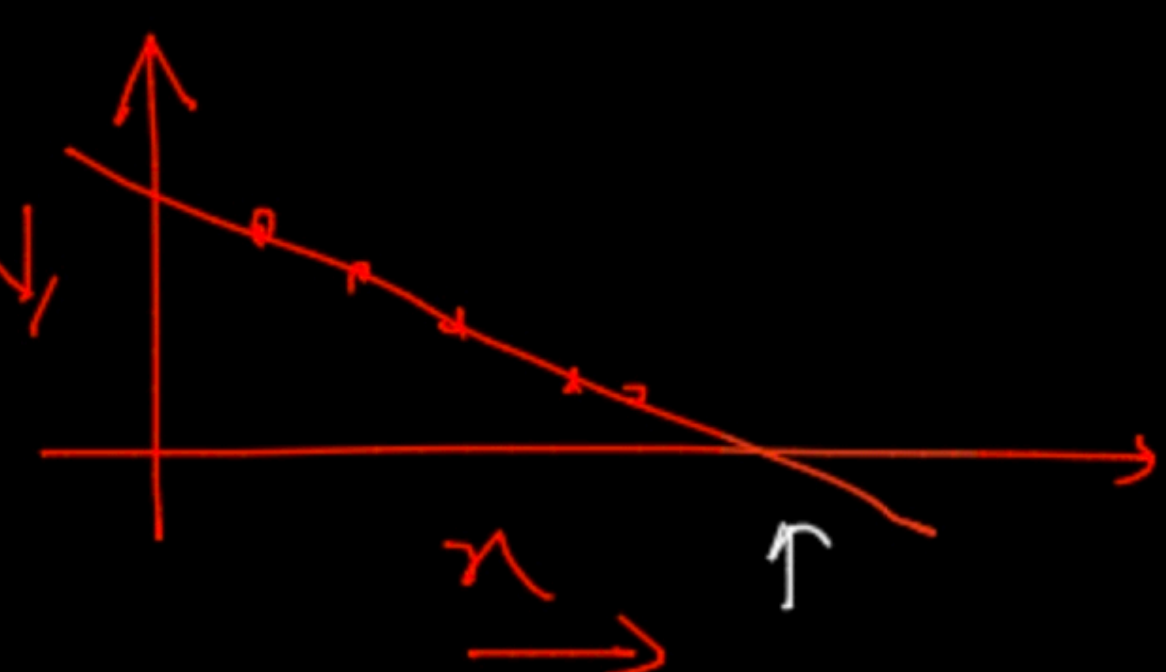


(x↑ y↑)

if $\rho_{X,Y} = -1$ ie strictly decreasing

⇓

$\rho = 0$ → No Relationship



(X, Y)

f_1 f_2 f_3 f_4

⇓

p - → heatmap

	f_1	f_2	f_3	f_4
f_1	1	0.4	0	0.5
f_2	—	1	—	—
f_3	0.6	—	1.	—
	—	—	—	1
f_4	—	—	—	1

f_2 f_3

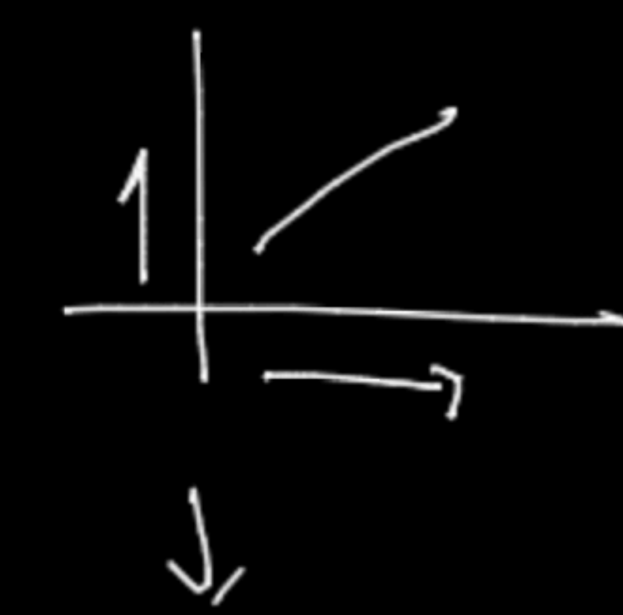
↑

Read → conclusion

correlation

Analysis

discount sales



(iii) Spearman's Rank correlation

X	Y	r_x	r_y
160	52	4	3
150	66	2	4
170	64	5	5
140	46	1	1
158	51	3	2

$$r = \rho_{r_x r_y} \Rightarrow \frac{\mathrm{cov}(r_x, r_y)}{\sigma_{r_x} \cdot \sigma_{r_y}}$$

$\Rightarrow$ discnt sales

'discnt'

$\Downarrow$

causation

correlation

(x causes y)

(x and y)